

Housing Returns over the Income Distribution

Claes Bäckman Walter D'Lima Natalia Khorunzhina

SAFE · Florida International University · Copenhagen Business School

Housing returns are equal across incomes, but their composition is not

Capital gains



rise with buyer income rank
P90 vs. P10: +1.36 pp per year

Rental yields



fall with buyer income rank
by roughly the same amount

Total return



flat or slightly declining
gradient ≈ 0

Where households buy explains the split. Borrowing constraints and inelastic supply explain where they buy.

Prior work infers housing returns from price indices and a uniform yield

Returns-to-wealth literature

Fagereng et al. (2020), Bach et al. (2020)

Aggregate house-price index
+ one rental yield for everyone

⇒ return differences can only come
from **price growth**

This paper

Danish registers, 1996–2022

Realized gain on *each* repeat sale
+ **rental yield** of *each* postcode

⇒ we can see whether yields *offset*
price growth

Housing is the dominant asset: 53% of gross wealth in Denmark.

We link every Danish repeat sale to the buyer's income rank

218,000

repeat sales, 1996–2022

605

postcodes, 98 municipalities

0–100

income rank within year ×
age cohort

Annualized, unlevered, real log capital gain on each property:

$$r_i^u = \frac{\ln P_{is} - \ln P_{ip}}{T_{is} - T_{ip}}$$

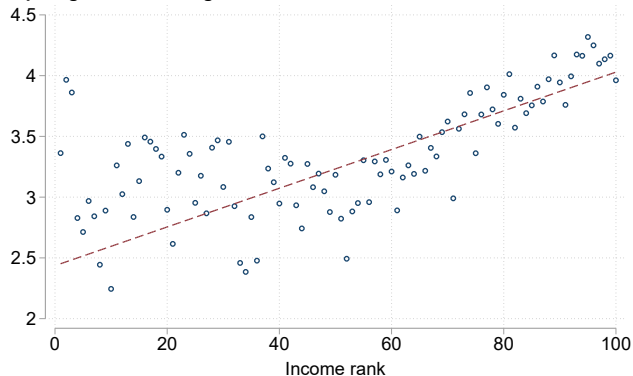
Rental yields imputed by postcode from 1999 rents in the housing register, validated against national accounts and 2015–22 private rental data.

Part 1: Capital gains

Higher-income buyers earn more price appreciation

P90 buyers earn 1.36 pp more capital gains per year than P10 buyers

Capital gains to housing



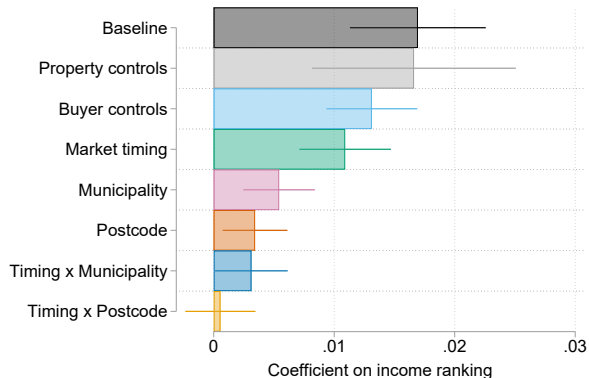
Slope: 0.017

pp per rank, s.e. clustered by municipality

$$80 \times 0.017 = \mathbf{1.36 \text{ pp}} \text{ per year}$$

Over a 10-year holding period the gap cumulates to **14.6%** of the purchase price.

Where households buy, not what they buy, accounts for the gap



Property and buyer controls

0.017 $\rightarrow$ 0.013

National market timing

0.013 $\rightarrow$ 0.011

Postcode fixed effects

$\rightarrow$ 0.003

Location accounts for **88%** of the decline (Gelbach decomposition), regardless of the order of controls.

Within a local market, the income gradient turns negative

Capital gains on income rank, with postcode and timing fixed effects

	Capital area	Capital + 3 cities	8 largest cities	7 cities excl. Cph.
Income rank	−0.004*** (0.001)	−0.003*** (0.001)	−0.003*** (0.001)	−0.004** (0.001)
Observations	93,372	74,702	106,098	51,153

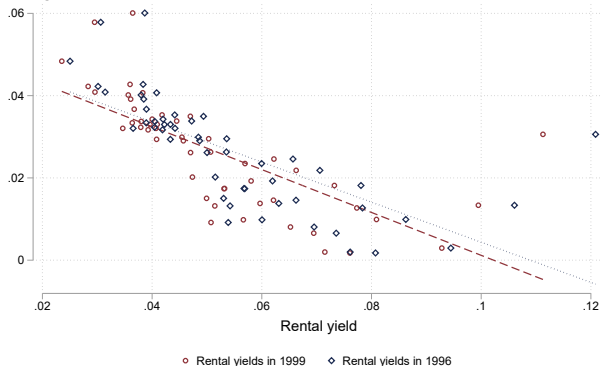
Higher-income buyers do not pick better properties inside a market. This is evidence against skill, information or sophistication as the source of the gap. The question becomes **why they buy in different markets.**

Part 2: Total returns

Capital gains are only half of the return to owning

Across municipalities, yields are lowest where prices grew fastest

Average return 1997-2022



Each dot is a municipality.

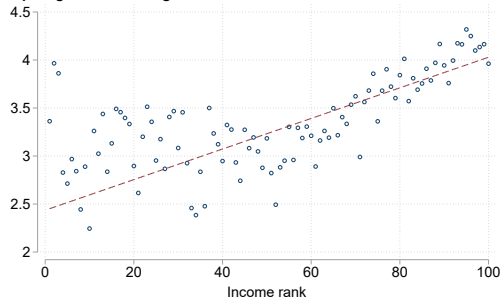
Rental yield on the x-axis
Average capital gain on the y-axis

The 1996 and 1999 cross-sections give the same pattern. High-priced markets combine high appreciation with low yields.

Rental yields fall with income as steeply as capital gains rise

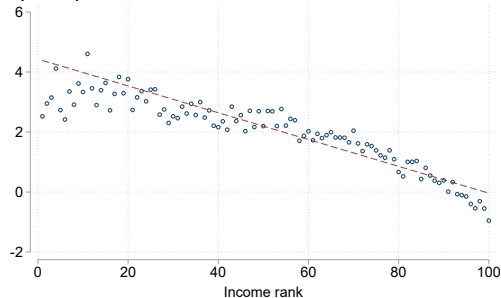
Capital gains

Capital gains to housing



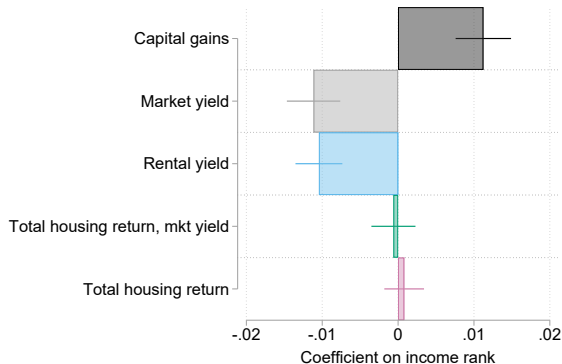
Net rental yield

Implicit imputed rents



Higher-income buyers concentrate in expensive markets, where rents are low relative to prices.

The two gradients offset: the total return does not rise with income



Capital gain

positive

Rental yield

negative, similar size

Total, additive

≈ 0

Total, IRR

-0.031

The IRR also nets out maintenance, property tax and transaction costs, which pushes the gradient below zero.

Only capital gains build wealth. Rental yields are consumed.

Total return to owning: $r^H = g + h$

Change in net worth:

$$\Delta W_{it} = g_{it} H_{it} + G_{it}^O + S_{it}^A$$

G^O : gains on other assets S^A : active saving

High-income buyers

return arrives as price appreciation that is lumpy, untaxed and stays on the balance sheet

Low-income buyers

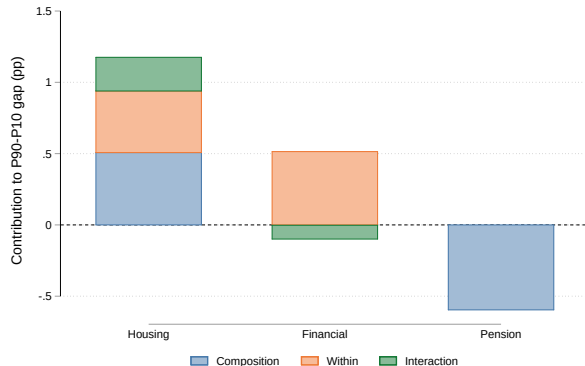
return arrives as a flow of housing services that supports consumption

Equal total returns therefore need not mean equal wealth accumulation.

Housing is the largest contributor to the P90–P10 gap in wealth returns

Decomposition following Kuhn et al. (2020), with Danish portfolio shares.

Housing adds **1.2–1.3 pp** per year to the P90–P10 gap (0.3 pp under the IRR)



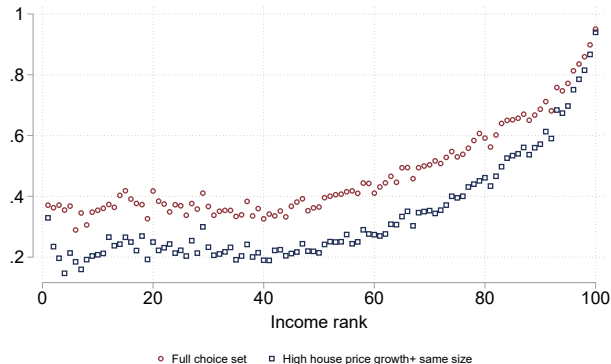
Under the additive measure this works through portfolio shares: housing is 48% of gross assets at P90 but 36% at P10.

Part 3: Why buyers sort

Constraints and supply keep lower-income buyers out of high-growth markets

Lower-income buyers can afford few homes in high-growth areas

Feasible transaction share



Feasible set: homes priced below

$$P^{\max} = \min(M^{LTV}, M^{PTI}) + W$$

80% LTV, 35% PTI, same-year transactions

All homes:

≈40% for bottom third, >60% for top third

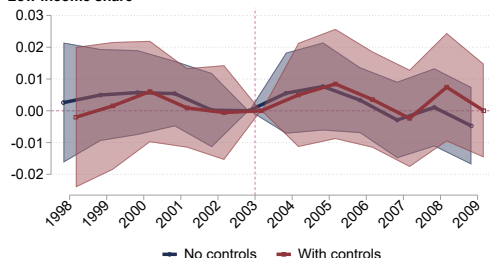
High-growth, same size:

≈20% at the 40th percentile

Easing credit in 2003 raised prices but did not change who buys

Share of low-income buyers

Low income share



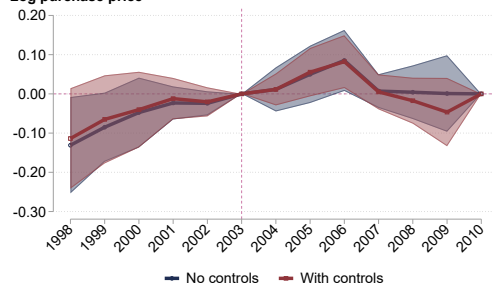
Share of low income buyers 1998-2002

All transactions: 7.841%. High-return municipalities.: 6.499%

Pre-trend F-test: $F(\text{No controls})=0.602$ ($p=0.699$), $F(\text{Controls})=0.526$ ($p=0.756$)

Log purchase price

Log purchase price



Pre-trend F-test: $F(\text{No controls})=5.101$ ($p=0.000$), $F(\text{Controls})=6.951$ ($p=0.000$)

Interest-only loans cut payments by about 20%. Take-up exceeded 60% at all incomes.
Effect on the low-income share: 0.003 (s.e. 0.003). Rules out increases above 0.9 pp.

Where supply is inelastic, credit is capitalized into prices

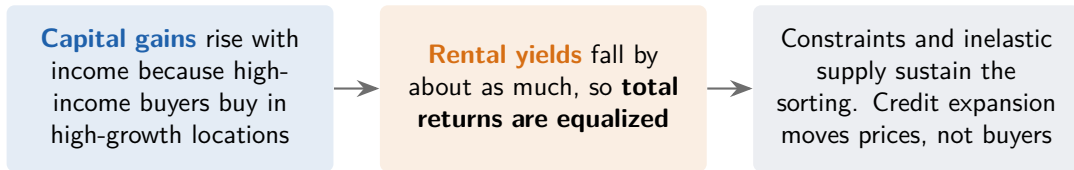


Share of middle-income buyers in high-growth municipalities after 2003:

Post × HighGrowth	+0.127**
Post × HighGrowth × inverse supply elasticity	−0.102***

Middle-income buyers entered where supply is elastic, and did not enter where it is tight.

Equal returns, unequal wealth



Housing matters for wealth inequality through the *composition* of its return, not its level.

Realized returns over 1996–2022, and a statistical decomposition rather than a causal estimate of location.

Thank you

baeckman@safe-frankfurt.de

Appendix

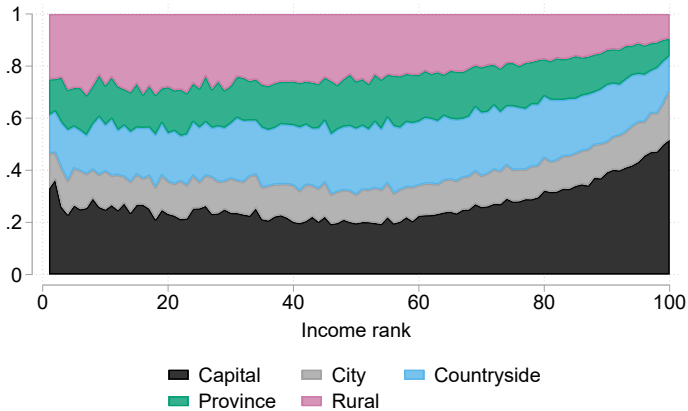
High-income buyers pay more, but buy similar properties

	All	Low income	Middle income	High income
Purchase price (DKK)	1,205,815	953,824	1,036,606	1,360,897
Annualized real capital gain (%)	3.6	3.2	3.1	4.0
Holding period (years)	8	8	8	8
Apartment share	0.31	0.35	0.24	0.34
Size (m ²)	112	101	112	113
Building age	56	59	55	56
Pre-purchase income (DKK)	324,452	125,490	220,140	427,337
Wealth rank	46	50	39	50
Age	41	45	38	42
Renovation indicator	0.34	0.26	0.29	0.39
Share of all transactions		0.09	0.36	0.55
N (repeat sales)	217,926	19,826	79,329	118,771

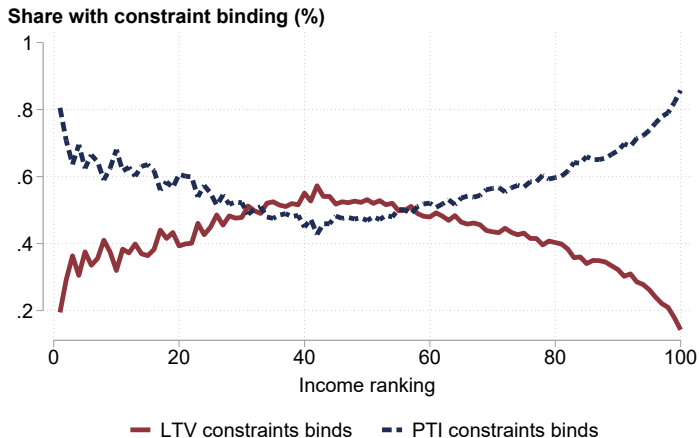
Income groups are terciles of the income rank. Repeat-sales sample, 1996–2022.

Buyers in the capital region become more common above the 60th percentile

Share of buyers by urbanization level



The payment-to-income limit binds for about half of buyers

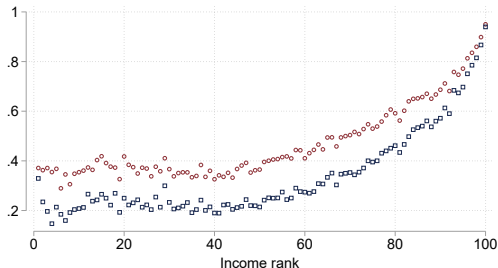


The LTV share is hump-shaped in income. Differences in maximum borrowing come mostly from PTI.

Choice sets look the same when restricted to the buyer's own region

Within country

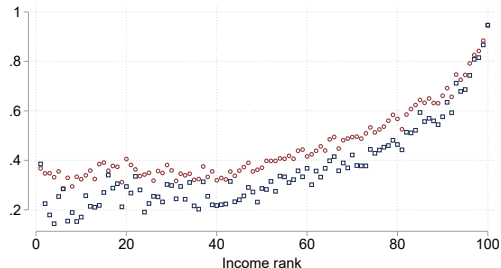
Feasible transaction share



○ Full choice set □ High house price growth + same size

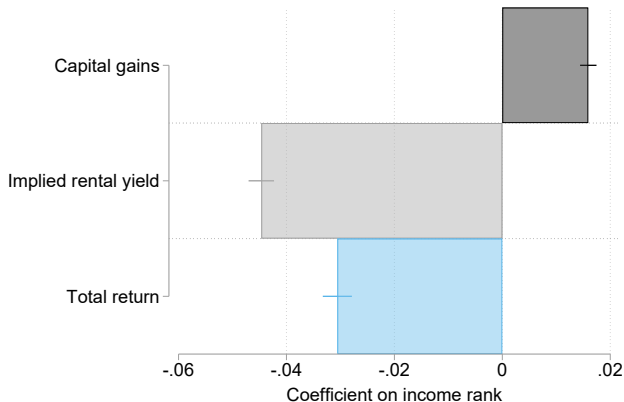
Within region

Feasible transaction share



○ Within region choice set □ High house price growth + same size

Under the IRR, the total-return gradient is negative in every specification



Maintenance 1.5% of value per year, 0.6% closing costs, 4% sale costs, observed property taxes.

The 2003 reform did not shift the income mix in high-growth areas

	(1) Low	(2) Middle	(3) High	(4) Low	(5) Middle	(6) High
Post $\times$ HighReturn	0.003 (0.003)	-0.008 (0.007)	0.013* (0.007)	-0.009 (0.021)	0.127** (0.048)	-0.174** (0.078)
Post $\times$ HighReturn $\times$ Supply Elast.				0.009 (0.015)	-0.102*** (0.036)	0.142** (0.058)
Observations	393,352	393,352	393,352	393,352	393,352	393,352

Municipality, year and region $\times$ year fixed effects, property and buyer controls. S.e. clustered by municipality.